AI FINANCIAL ASSISTANT

Project Documentation

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1. Abstract

The AI Financial Assistant is a browser-based web application that helps users make quick, simple financial estimates. From values the user enters, it shows a financial profile summary, a loan eligibility estimate, the monthly EMI of a loan, and a credit score category. A fifth module, AI Financial Tips, sends the entered income, EMI and credit score to a separate backend service and displays the text it returns. The tools are built with HTML, CSS and JavaScript, and the calculations and rules run in the browser. In the tested build, the four tools that run in the browser worked as described. The AI Financial Tips module could not be verified because its backend service was not part of the submitted files.

2. Introduction

Taking a loan, or deciding whether one is affordable, involves several figures: monthly income, existing EMIs, credit score, loan amount and tenure. Working these out by hand is slow and error-prone. This project puts the common checks on one page with a dashboard, one tool per task, and short results that explain the outcome. It is an educational project; every result is an estimate based on simplified rules and is not a bank decision.

3. Problem Statement

Many people do not have a quick way to check how a loan fits their income and credit profile before they approach a lender. Bank calculators are usually separate from eligibility tools, and the results are hard to understand. The proposed assistant brings the main checks together on one page and explains each result in plain language.

4. Objectives

- Present a single dashboard with links to every financial tool.
- Generate a financial profile summary with the EMI-to-income ratio, an EMI status and a credit category.
- Estimate loan eligibility from age, income, credit score and existing EMI, and list the conditions that were not met.
- Calculate the monthly EMI, total interest and total payment for a given loan amount, interest rate and tenure.
- Classify a credit score into a simple category.
- Validate inputs and show clear feedback for missing or invalid values.
- Request financial tips from a backend service and handle the case where it is unavailable.

5. Scope

Core features: the dashboard and navigation, Financial Summary, Loan Eligibility Checker, EMI Calculator and Credit Score Analyzer, all of which run in the browser, plus the AI Financial Tips module, which depends on a backend service. User accounts, saved history, charts, a real prediction model and bank integration are not implemented and are listed as future enhancements.

6. Technology Stack

- Frontend: HTML5 for structure (index.html); CSS3 for the dark theme, grid and flexbox layout, sticky navigation, hover effects and responsive media queries (style.css).

Logic: JavaScript (script.js) using DOM event listeners, template literals, Number() conversion, fetch with async/await, and toLocaleString("en-IN") for Indian number formatting.

Backend: none included. The AI Financial Tips module sends a POST request to http://localhost:5000/api/financial-insights and expects a JSON reply with success and insights or message fields.

Database: none. No data is stored.

AI/insights: the Summary, Eligibility and Credit tools use fixed rules written in script.js. The tips text comes from the external backend, whose model or logic was not available for this report.

Test environment: Chromium 153 run headless through Playwright for Python on Linux, opening the project files directly in the browser (25 October 2026).

7. Requirements

A laptop, desktop or mobile device with a modern web browser.

The three project files kept in one folder: index.html, style.css and script.js.

No installation is needed for the Summary, Eligibility, EMI and Credit tools.

For AI Financial Tips: a backend service running at http://localhost:5000 that provides the /api/financial-insights endpoint.

8. System Modules

Navigation and Dashboard: a sticky top menu and four feature cards that scroll to each tool.

Financial Summary: takes income, existing EMI, credit score, requested loan amount and tenure; shows the EMI-to-income ratio, EMI status, credit category and an eligibility estimate.

Loan Eligibility Checker: takes age, income, credit score, existing EMI, loan amount and tenure; shows Eligible or Not Eligible, with the reasons if not eligible.

EMI Calculator: takes loan amount, annual interest rate and tenure in years; shows the monthly EMI, total interest and total payment.

Credit Score Analyzer: takes a credit score and shows its category with a short tip.

AI Financial Tips: takes income, existing EMI and credit score, sends them to the backend and displays the returned text, or a connection error.

9. Workflow

1. The user opens index.html and uses the menu or a dashboard card to reach a tool. 2. The user enters values and clicks the tool's button. 3. The script converts the inputs to numbers and checks them; if any value is missing or invalid, a warning is shown. 4. For the Summary, Eligibility, EMI and Credit tools, the script applies its rules or formula in the browser. 5. For AI Financial Tips, the script sends the values to the backend and waits for the reply. 6. The result is displayed in a result card below the form, with a note that it is an educational estimate.

10. Data Design

The project has no database and does not store user data. Each tool reads its own input fields, and results are shown on the page and are lost when it is reloaded. The fixed rules used by the tools are listed below.

Eligibility conditions (Loan Eligibility and Summary): age at least 18 (Loan Eligibility only); credit score at least 650; monthly income at least ₹15,000; existing EMI no more than 50% of monthly income.

Credit categories: 750 and above is Excellent; 700 to 749 is Good; 650 to 699 is Fair; below 650 is Needs Improvement.

EMI status (Summary): EMI-to-income ratio up to 30% is Manageable; above 30% and up to 50% is Moderate; above 50% is High.

EMI formula: EMI = P × r × (1 + r)^n ÷ ((1 + r)^n – 1), where P is the loan amount, r is the annual rate ÷ 12 ÷ 100 and n is the tenure in years × 12. At 0% interest, EMI = P ÷ n.

Request sent to the backend: a JSON object with income, emi and creditScore.

11. Implementation

The page has one index.html with a section for each tool, and each section has its own inputs, button and result card (the result card is hidden until used). script.js attaches a click listener to each button. Each listener reads the fields, validates them (for example, income, loan amount and tenure must be greater than zero, and EMI cannot be negative), and writes an HTML result into the result card. Amounts are formatted in the Indian style, such as ₹5,00,000. The AI Financial Tips listener shows a loading message, calls the backend with fetch, and displays either the returned insights, the backend's error message, or a Backend Connection Error message if the request fails. style.css provides the dark theme, two-column dashboard on wide screens and a single column on narrow screens, and a menu that scrolls sideways on small screens.

12. Testing

The tests were run on the project files in a headless Chromium browser. Each row shows what was entered and what the page displayed.

Test case	Expected result	Actual result
Page load and navigation	Page opens without script errors; menu and cards are present	Title “AI Financial Assistant”; 0 script errors; 5 menu links and 4 dashboard cards
Eligible profile (age 28, income ₹50,000, score 750, EMI ₹10,000, loan ₹5,00,000, 5 years)	Estimated Eligible	Estimated Eligible, showing loan ₹5,00,000, tenure 5 years and score 750
Ineligible profile (age 17, income ₹12,000, score 600, EMI ₹8,000)	Not Eligible with reasons	Estimated Not Eligible; all four reasons listed (age, credit score, income, EMI)
Boundary values (age 18, income ₹15,000, score 650, EMI ₹7,500)	Estimated Eligible	Estimated Eligible
EMI just above 50% of income (₹20,001 on ₹40,000)	Not Eligible; EMI reason only	Estimated Not Eligible; only the EMI reason listed
Loan form left empty	Validation feedback appears	“Please enter valid details” warning shown
EMI: ₹5,00,000 at 8.5% for 5 years	EMI about ₹10,258	EMI ₹10,258.27; interest ₹1,15,495.94; total ₹6,15,495.94
EMI: ₹1,20,000 at 0% for 1 year	EMI ₹10,000	EMI ₹10,000; interest ₹0; total ₹1,20,000
EMI form empty; negative interest rate	Validation feedback appears	“Please enter valid values” shown in both cases
Credit scores 780, 720, 670, 600	Excellent, Good, Fair, Needs Improvement	Categories matched for all four scores

Test case	Expected result	Actual result
Credit score boundaries 750 and 649	Excellent; Needs Improvement	Excellent for 750; Needs Improvement for 649
Credit form left empty	Validation feedback appears	“Please enter a valid credit score” shown
Summary: income ₹60,000, EMI ₹12,000, score 760	Ratio 20%, Manageable, Excellent, Eligible	Ratio 20.0%; Manageable; Excellent; Estimated Eligible
Summary: income ₹14,000, EMI ₹9,000, score 610	Ratio 64.3%, High, Needs Improvement, Not Eligible	Ratio 64.3%; High; Needs Improvement; Estimated Not Eligible
Summary form left empty	Validation feedback appears	“Please enter valid details” warning shown
AI Financial Tips form left empty	Validation feedback appears	“Please enter valid details” warning shown
AI Financial Tips with valid values, backend not running	Tips if the backend is running; an error message if it is not	“Backend Connection Error” shown; the tips themselves were not tested
Mobile width (390 px)	Layout fits without sideways scrolling	No horizontal overflow detected

Not tested: the “Moderate” EMI status band (ratio above 30% up to 50%) and the Good-category boundary at 700. The AI Financial Tips result could not be tested without the backend service.

13. Results and Screenshots

The figures below are screenshots of the running application captured during testing. For the screenshots, the sticky menu was set to scroll with the page so that it does not cover the forms; the application's files were not changed.

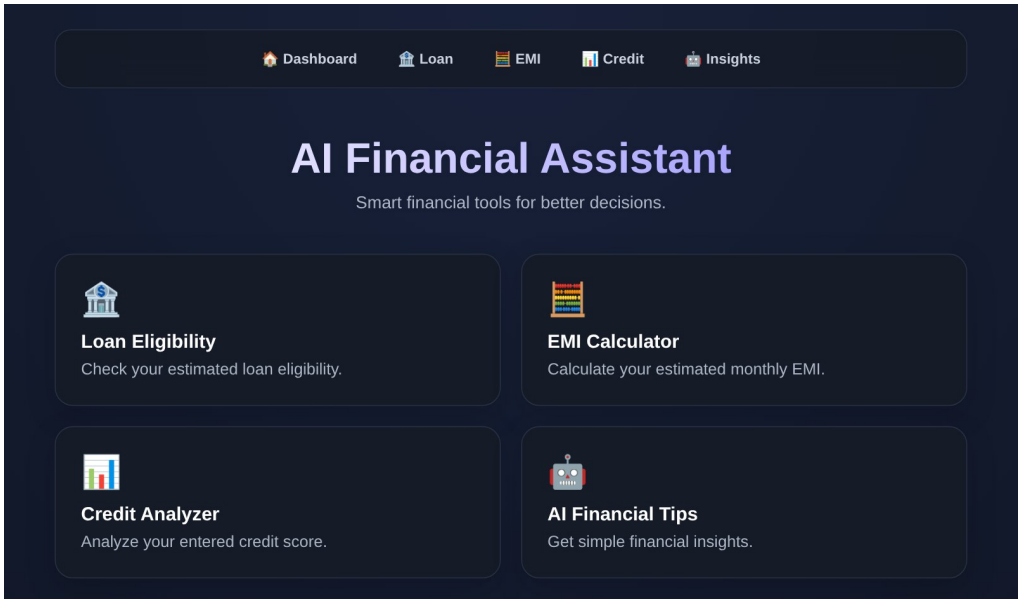


Figure 1—Dashboard

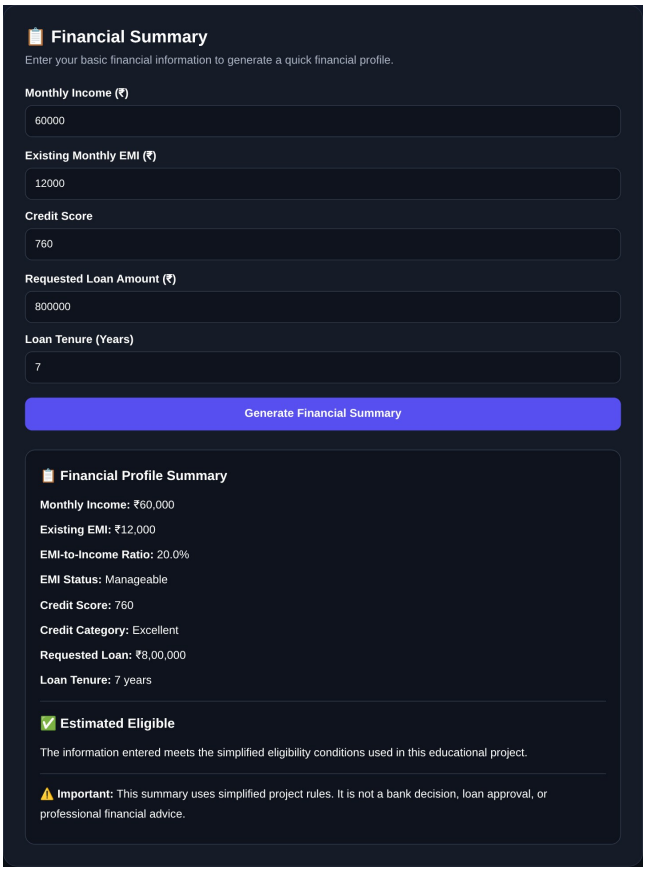


Figure 2—Financial Summary result

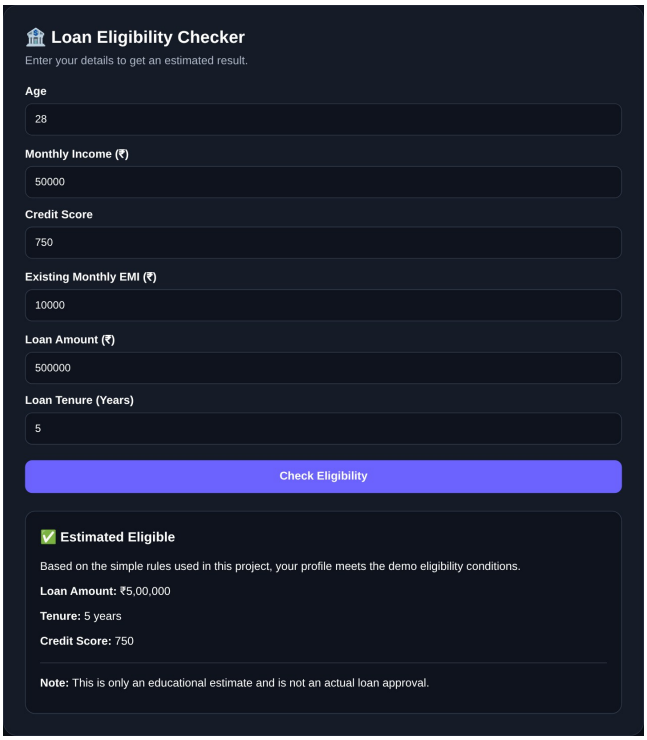


Figure 3—Loan Eligibility: eligible result

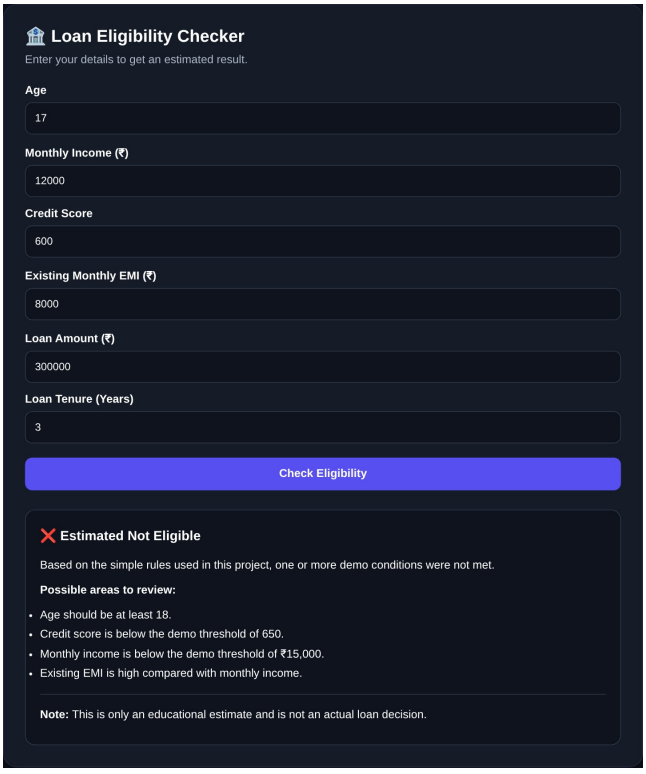


Figure 4—Loan Eligibility: not-eligible result with reasons

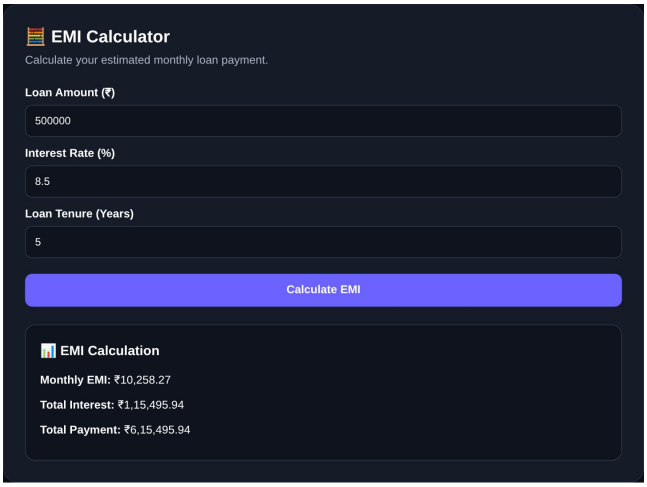


Figure 5—EMI Calculator result (₹5,00,000 at 8.5% for 5 years)

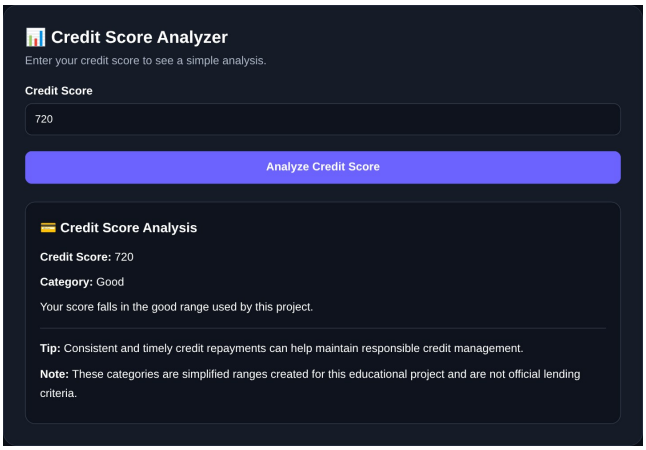


Figure 6—Credit Score Analyzer result

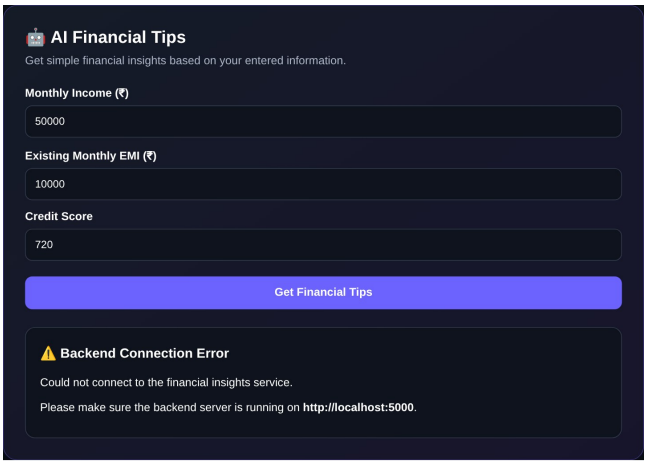


Figure 7—AI Financial Tips: connection message shown when the backend is not running

14. Security, Privacy, and Limitations

The four browser-based tools do not send or store any data. The AI Financial Tips module sends the entered income, EMI and credit score to http://localhost:5000, which uses plain HTTP and is meant for a local development server. The following limitations apply to the tested build:

There is no login or user accounts, and no data is saved.

The text returned by the backend is inserted into the page as HTML without cleaning. A deployed version should sanitize this text or insert it as plain text, and use HTTPS.

The backend service was not available, so AI Financial Tips could not be tested and the quality of its tips is unknown.

Loan amount and tenure are collected and displayed by the Loan Eligibility Checker, but they do not affect the eligibility decision. The decision uses only age, credit score, income and existing EMI.

Input checks are basic: values must be positive (EMI can be zero), but there are no upper limits, such as a maximum age or a credit score range.

Thresholds such as 650 and ₹15,000 are simplified project rules and are not real lending criteria.

All results are educational estimates and are not financial advice or a loan decision.

15. Future Scope

Provide and document the backend service, with a real model or a tested rule set for the insights.

Use loan amount and tenure in the eligibility decision, for example by comparing the new EMI with income.

Add charts, such as a repayment breakdown of principal and interest.

Add stronger input validation, with limits and clear messages for each field.

Add user accounts and saved history, with proper security for stored data.

Move to HTTPS and sanitize all backend output.

Add accessibility improvements and further mobile testing.

16. Conclusion

The AI Financial Assistant puts a financial summary, loan eligibility check, EMI calculator and credit score analyzer on one page. In testing, these four browser-based tools gave correct and consistent results, including at their boundary values, and showed clear messages for empty or invalid input. The AI Financial Tips module depends on a backend service that was not available, so it remains to be tested. With a working backend and the improvements listed in Future Scope, the project can become a more complete financial awareness tool.

17. References

MDN Web Docs, HTML: https://developer.mozilla.org/en-US/docs/Web/HTML

MDN Web Docs, CSS: https://developer.mozilla.org/en-US/docs/Web/CSS

MDN Web Docs, JavaScript: https://developer.mozilla.org/en-US/docs/Web/JavaScript

MDN Web Docs, Fetch API: https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API

Playwright for Python (used for testing): https://playwright.dev/python/